

Verification and Progress on Purity of Pure

A machine-checked report on Goodman’s project notes

27 July 2026

Purpose. This report records what we have verified in Jeremy Goodman’s July 2026 project notes on the consistency of Purity of Pure with Bacon’s logical combinatorialism. It has two aims. First, it separates the claims that have been verified exactly from claims that require qualifications, repairs, or further formulation. Second, it states the new theorems proved during the formalization that genuinely narrow the open consistency problem.

Bottom line. Goodman’s central consistency question remains open. But the formalization has substantially changed its shape. The positive model program is now reduced to constructing one enumeration of the pure unary operators. The enumeration must respect Bacon’s substitution action, and its range must contain every unary operator definable using the enumeration itself. All remaining soundness obligations for this reduction have been discharged. Separately, a HOL-ZF model validates PP and the background principles listed below together with purity of identity, negation, constant truth, and constant falsity. Consequently every subcollection of Goodman’s principles whose logical-purity instances are confined to those four formulas is consistent in Bacon’s proof system. This does not settle the question, because Goodman’s intended theory makes every closed logical term pure at every type. Conversely, any such model must falsify global L2 or global TU. We have also verified every determinate object-language result in the notes, corrected a modal gap in T3, corrected a false countability argument in M5, proved the T9 cardinal dichotomy from its explicit counting assumptions, and isolated the exact qualifications required by several model-theoretic claims. Every determinate claim in the notes has now been proved, refuted and repaired, or assigned the explicit qualification on which it is true.

1 Formal setting and verification standard

The formal development represents Bacon’s typed language and proof system directly in Isabelle/HOL. It includes Bacon’s system H , classical and Boolean identity principles, the Rule of Equivalence, vector Equivalence, the definition of necessity by identity with truth, and the additional purity/fundamentality vocabulary used in Goodman’s notes. The development distinguishes object-language derivability from validity in a particular model. This matters throughout: a semantic argument valid in Bacon’s appendix model is not thereby a

derivation in the theory, and an Isabelle theorem with explicit premises does not show that those premises hold in the intended model.

The complete audit checks 96 principal results. A fresh build passes. None of these proofs contains an admitted lemma, an untrusted proof step, or an undischarged assumption beyond the assumptions stated in the theorem itself. The type restriction carried by the general cardinal theorems in T9 is part of their formally checked statement, not a further mathematical premise.

2 Background definitions and two important distinctions

At proposition type, Goodman's definable surrogate for fundamentality is

$$\text{fun}'(p) \stackrel{\text{def}}{=} \forall X Y ((\text{Pure}(X) \ \& \ \text{Pure}(Y) \ \& \ Xp = Yp) \rightarrow X = Y).$$

Let G be the pure reversible unary operators, and let

$$X \approx Y \stackrel{\text{def}}{=} \exists Z \in G (X = Y \circ Z).$$

Weak L2 says that if X and Y are pure, p and q are fun' , and $Xp = Yq$, then $X \approx Y$.

Two distinctions uncovered by the formalization are central.

Recombination versus Exhaustion. Unary Recombination proves a pointwise modal identity principle. It does not, by itself, permit necessitating the universal closure required by the usual QSS derivation. Zeroary Exhaustion supplies exactly that missing step. With unique proposition-level fundamentality, the repaired package derives the existence of a fun' proposition.

Conditional model-theoretic results. A formally verified theorem may still have explicit premises concerning Bacon's function domains, invariance, or QSS. We therefore distinguish a verified conditional theorem from a proof that all of its premises hold in Bacon's particular appendix model.

3 Audit of the object-language theorems

Item	Status	Result
T1	Verified	Under zeroary Exhaustion, every pure proposition is truth or falsity. Consequently the biconditional operators are identity and negation, and WI collapses to Inv.
T2a	Verified	fun' is closed under every pure reversible operator, hence under negation.
T2b	Verified	Truth and falsity are not fun' . A fun' witness is distinct from truth, falsity, and its negation.
T2c	Verified, stronger	Every pure proposition is possibly identical to a fun' witness. The proof does not use PP.
T2d	Verified, stronger	Possible purity of the witness follows without Persistence, although Goodman's displayed route uses it.
T2e	Verified	The proposition that the witness is noncontingent is false but possible.
T2f	Verified conditionally	The six displayed propositions are pairwise distinct under the explicit $\text{fun}'(r)$ antecedent. The antecedent is essential.
T3	Repaired	The advertised proof has a modal gap: it reaches possible identity where identity is required. The theorem follows after adding zeroary Exhaustion, or from a weaker pure-identity rigidity principle. A two-world modal abstraction verifies that the advertised intermediate principles do not by themselves close the gap.
T4	Verified, stronger	For each closed pure $C : t \rightarrow (t \rightarrow t)$, $\sim \text{fun}'_{t \rightarrow t}(C(r))$. The proof uses only the purity schema and application closure; neither PP nor $\text{fun}'(r)$ is needed.
T5	Verified conditionally	The assumptions displayed in the notes, together with $\text{fun}'(r)$, yield a fun' proposition distinct from r and $\sim r$. This does not establish consistency of the antecedent.

Item	Status	Result
T6	Verified	All four contradiction routes are formalized: L2+Inv, L2+Wl, L2+TU, and strong-L2+RS. Recombination, the proposition-type instance of Exhaustion, and unique fundamentality derive, rather than assume, the required fun' witness where appropriate. The advertised T6-Wl master claim is also derived directly from its stated stock, independently of explosion.
T7a	Verified	For the liar family, the identity-indexed member is false, while some member indexed by a pure reversible operator is true.
T7b	Underspecified	The notes do not give the kind-level diagonal term, its binders, or precise definitions of “fixed” and “shifted” point. There is no unique theorem to encode.
T8	Verified	The five base kinds are pairwise distinct; L2 gives kind uniqueness; and the assumptions displayed in the notes prove 31 pairwise-distinct pure unary operators and 31 pairwise-distinct propositions.
T9	Verified at the counting level	The PC specification makes the map from sets of kinds to pure operators injective. Representation of the members of each kind by members of G gives an injective code of that kind into G . These results imply $\mathcal{P}(K) \leq K \times G .$ Classical cardinal arithmetic then gives $K \text{ finite} \quad \vee \quad \mathcal{P}(K) \leq G .$ Instantiating the abstract types uniformly across Goodman's full typed object language remains separate.

4 Audit of the model-theoretic claims

Item	Status	Result
M1	Exact identification; exclusion conditional	At proposition type, the pure propositions in Bacon's model are exactly the two invariant propositions, and the operator testing for membership in that class is the noncontingency operator. The exact footnote-59 liar is typed, beta-reduced, and proved pure from PP plus Purity of Fun; given QSS, this yields a contradiction. At the next type, the infinitary join exists in the full function domain and has exactly the pure unary operators as its extension. It interprets Pure at that type. PP holds there exactly when this operator is itself in the next pure stock. The current formalization does not prove directly that it is absent from the next pure stock in Bacon's exact generic model. It proves that absence only from the additional PP-diagonal and QSS assumptions. Completeness of the function domain therefore supplies the join, but does not by itself settle its purity.
M2	Verified	Goodman's construction $T \mapsto G_T$ is a bijection between arbitrary sets of propositions and invariant unary operators. Cantor's theorem yields more invariant operators than propositions, so evaluation at one fundamental proposition cannot be injective on all invariants. The invariance reading of purity therefore violates QSS.
M3	Verified relative to the pure operators	Relative to the chosen stock of pure operators, fun' is equivalent to freeness against nonzero pure laws. Such propositions have both extreme views. A countable Boolean stock has a free generator, while the resulting fun' class is topologically meager in the product topology.

Item	Status	Result
M4	Conditional model-theoretic theorem	Preimage heredity is proved. The lifted-branch witness is fun' and outside the reversible orbit under explicit assumptions: necessitated QSS, function-space membership and invariance of every operator in the pure stock, and presence of identity, zero, and one. The theorem has not yet been instantiated for Bacon's stock of operators denoted by closed terms containing only logical vocabulary. The multiple-fundamental "wide Fun" argument remains prose-level.
M5	Verified, repaired, and partly open	The operator $f = G_T$ obtained by exchanging s and s' is an invariant involution violating TU, WI, and Inv. The notes' subsequent countability argument is false: both the orbit and the displayed world-indexed family are countable. We replace it with a uniform diagonal construction. For every candidate R , it produces a two-element pair outside every view of R , whose proper views are all empty or universal. The induced invariant nonidentity transposition fixes R . If it and identity are both included among the pure operators, R ceases to be fun' , proving the corrected pre-rebuild QSS obstruction. The full rebuilt model and the proposed generalization of the collision method remain separate questions. The displayed collision calculation itself is verified: the closed operator

$$\lambda p (p \leftrightarrow \text{NC}(p))$$

takes the same value at $\text{NC}(r)$ and truth, although $\text{NC}(r) \neq \text{truth}$, whenever $\text{fun}'(r)$. Extending this argument uniformly to merely existentially given invertibles remains open.

Item	Status	Result
M6	Conditional model-theoretic theorem	Under the same explicit assumptions—necessitated QSS, membership of every pure operator in Bacon’s function domain, and invariance—distinct substitutions are separated by a fun' proposition. With identity, zero, and one, a strict-inclusion fun' pair blocks the indicated joint assignment. Single-coordinate arbitrary realization fails independently by countability. These results have not all been instantiated for Bacon’s stock of operators denoted by closed terms containing only logical vocabulary.
M7	Verified, with an open instance	The Tarski diagonal lies outside the range of the enumerated closed terms. For arbitrary invariant operators of Goodman’s form G_T , every proposition is reachable from r if and only if the orbit map $i \mapsto i \cdot r$ is injective. Whether Bacon’s glued r has this property is construction-sensitive and open.
10.1	Verified, qualified	Bacon’s rebuilt-family theorem is proved in a set-theoretic formalization of his appendix model. At type Ind , equality is literal identity, and the theorem’s hypotheses therefore force the family to be constant; the theorem cannot supply a family of distinct individual denotations.

5 Corrections and clarifications to the notes

5.1 The modal gap in T3

The advertised argument uses necessitated QSS and Persistence to move from a possible fundamental role to an injectivity claim. The formal derivation reaches a statement of the form

$$\Diamond(Y = Z),$$

where the desired conclusion requires $Y = Z$. No rule of H licenses that transition. The gap can be repaired in either of two principled ways.

1. Zeroary Exhaustion collapses pure propositions to truth and falsity, which supplies the required rigidity.
2. One may assume only the weaker principle that possible identity between pure operators implies identity.

The unrestricted version of the second principle is false. Thus the repair must remain restricted to the pure case.

5.2 The countability error in M5

The notes propose choosing a pair $\{s, s'\}$ outside the orbit of r because the orbit is countable and, it is said, there are uncountably many world-indexed pairs. But Bacon's worlds are finite sequences of natural numbers, hence countable. The displayed family of pairs is therefore countable as well. Countability alone proves nothing.

The replacement is direct diagonalization. Enumerate the views of R by the worlds. Define a proposition s_R whose truth at the singleton $[n+2]$ disagrees with the corresponding singleton's membership in the n -th view of R , and let

$$s'_R = s_R \cup \{\langle \rangle\}.$$

Membership above a nonactual world depends only on the final symbol of its sequence. Hence every nonactual view of s_R and s'_R is empty or universal, while both members differ from every view of R . Exchanging their membership in the set T_0 of true propositions yields an invariant nonidentity operator $f_R = G_T$ with

$$f_R(R) = R.$$

If f_R and identity are both included among the pure operators, evaluation at R is not injective. This is exactly the pre-rebuild QSS obstruction the M5 strategy needs.

5.3 The scope of several semantic claims

The M4 and M6 theorems are genuine mathematical results, but at present they are conditional. They assume that every operator counted as pure belongs to Bacon's function

domain and is invariant, together with necessitated QSS and, for some results, the presence of identity and the two constant operators. The proofs do not yet instantiate these assumptions for Bacon’s complete stock of operators denoted by closed terms containing only logical vocabulary. We therefore do not count them as unconditional verifications of the corresponding claims about Bacon’s particular glued model.

6 New theorems advancing the consistency question

The audit was not merely defensive. The formalization has produced several theorems not stated in the original notes. These results materially narrow both the positive and negative programs.

6.1 Recombination, Exhaustion, and the existence of a fun' proposition

Theorem 1. *Unary Recombination yields the pointwise QSS core. Adding zeroary Exhaustion yields the necessitated universal form needed for QSS. Together with unique proposition-level fundamentality, the resulting stock proves that a fun' proposition exists.*

This theorem removes the previously assumed existential fun' premise from the premises used in T6. It also makes explicit exactly where Exhaustion, rather than Recombination, enters the object-language argument.

6.2 An exact semantic consequence of T6

Theorem 2. *Any model validating Bacon’s proof theory, Recombination, the proposition-type instance of Exhaustion, unique proposition-level fundamentality, and PP must falsify global L2 or global TU:*

$$\sim \text{GValid}(\text{L2}) \quad \vee \quad \sim \text{GValid}(\text{TU}).$$

Moreover, some world in the constructed model falsifies one of the two closed formulas.

Here $\text{GValid}(A)$ means that the closed formula A is true at every world of the model. This formalizes the central dilemma in Goodman’s notes. It does not select a disjunct. A positive model may exploit either cross-kind collisions on fun' inputs or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping.

6.3 Adding purity of constant truth and constant falsity

Let $C_{\top}$ and $C_{\perp}$ be the unary operators

$$C_{\top}(p) = \text{truth} \quad \text{and} \quad C_{\perp}(p) = \text{falsity}.$$

We now make these two closed logical operators pure, in addition to identity and negation. Closure of purity under application gives truth and falsity at type t , the four unary operators at type $t \rightarrow t$, and the proposition classifying those four unary operators at type $(t \rightarrow t) \rightarrow t$.

Theorem 3 (The four-logical-operators fragment). *There is a set-theoretic HOL-ZF tree model with Henkin domains that validates all of the following:*

1. *purity of identity, negation, $C_{\top}$, and $C_{\perp}$;*
2. *PP;*
3. *every instance of closure of purity under application;*
4. *zeroary and unary Recombination and Exhaustion;*
5. *Modalized Functionality at arbitrary types; and*
6. *unique fundamentality at proposition type and the absence of fundamental entities at every other type.*

The collection consisting of exactly these principles is a subcollection of Goodman's principles and is consistent in Bacon's proof system.

Let T_G be Goodman's full collection of principles, and let $\text{LP} \subseteq T_G$ be its logical-purity schema. Put

$$\text{LP}_4 = \{ \text{Pure}(I), \text{Pure}(N), \text{Pure}(C_{\top}), \text{Pure}(C_{\perp}) \},$$

where I is identity and N is negation. The bridge theorem establishes, for every U ,

$$(IC) \quad U \subseteq T_G \quad \text{and} \quad U \cap \text{LP} \subseteq \text{LP}_4 \quad \implies \quad \text{Con}_{\text{CEV}^+}(U).$$

No finiteness assumption on U is required. This does not settle Goodman's question, because the logical-purity schema in the intended theory includes every closed logical term, including terms at higher types. The next planned extensions add purity of K , then purity of conjunction; a complete result requires all closed logical terms at all types.

Isabelle verification note. The model and consistency theorems are:

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pp_t_logical_constants_fragment_PP_gvalid
pp_logical_constants_fragment_PP_axioms_consistent
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The inclusion in Goodman's principles and the general bridge theorem are:

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pp_logical_constants_fragment_PP_axioms_subset_fresh_goodman
fresh_goodman_logical_constants_only_consistent
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6.4 Reduction of the positive program to one equation

We constructed a set-theoretic formalization of Bacon's substitution model and verified directly:

- every axiom of Bacon's base proof theory;
- the vector Equivalence rule;
- beta and eta compatibility;
- the Boolean, quantifier, identity, and modal clauses used by the central theory;
- the proposition-type instance of Exhaustion for every relevant choice of pure entities.

Let E be an enumerator of unary operators, represented in the language by a term of type $\text{Ind} \rightarrow (t \rightarrow t)$. It must commute with taking the view from any substitution, as Bacon's function-domain condition requires. Let C_E be the class of unary operators denoted by closed expressions built from the logical vocabulary, the natural-number indices, E , and application.

Theorem 4 (Sufficient condition for a model). *If E has the preceding properties and satisfies*

$$(SE) \quad C_E = \{E(n) : n \in \mathbb{N}\},$$

then the resulting model validates Bacon's base theory, Recombination, the proposition-type instance of Exhaustion, unique proposition-level fundamentality, and PP at the target type. Hence (SE) gives an affirmative answer to Goodman's question.

Theorem 5 (Equivalent formulation). *The inclusion*

$$\{E(n) : n \in \mathbb{N}\} \subseteq C_E$$

is automatic. Thus (SE) is equivalent to the assertion that every well-typed unary expression built in the indicated way has the same denotation as $E(n)$ for some n .

This is the cleanest current formulation of the positive problem. No independent soundness premise remains. The residual task is not to construct an arbitrary set-theoretic enumeration, but an enumeration compatible with Bacon's substitution action whose range contains every operator definable using the enumeration itself.

6.5 Diagonal restrictions on any solution

Theorem 6 (Restriction on definable diagonal predicates). *If E satisfies (SE), then there is no expression $P : \text{Ind} \rightarrow t$, built in the indicated way, that diagonalizes against the enumeration at the actual world:*

$$P(n) \leftrightarrow \sim \text{Truth}(E(n)(P))$$

in the required sense. In particular, no definable predicate can assign different truth values to all indices in the naive Cantorian way.

This explains why a straightforward internal enumeration diagonal does not immediately refute (SE): compatibility with Bacon's substitution action collapses the obvious index predicates.

We also formalized a more flexible diagonal $D_R(E, H, R)$. It treats two representations as equivalent according to a relation R , rather than requiring literal identity. The construction is a closed term containing only logical vocabulary; consequently it is pure whenever its parameters H and R are pure.

Theorem 7 (Generalized diagonal constraint). *Assume (SE). Suppose R cannot distinguish propositions that agree in truth value at the actual world, and H distinguishes the relevant indices. Then some proposition $H(q)$ has two representations that are not R -related.*

One natural choice says that F and G are related at q when

$$F = G \circ \psi \quad \text{for some bijection } \psi \text{ with } \psi(q) = q.$$

This relation is itself expressible using only logical vocabulary and cannot distinguish propositions agreeing in actual truth value. It relates every pair obtained by a reversible change of input that fixes q , eliminating that entire class of prospective counterexamples. What remains is either to show that it relates all representations of some suitable $H(q)$, or to construct two representations that it does not relate.

6.6 A uniform M5 obstruction

Theorem 8 (Uniform version of the M5 construction). *For every candidate proposition R , there is a pair $\{s_R, s'_R\}$ disjoint from the set of all views of R . Every nonactual view of either member is empty or universal. The associated invariant transposition fixes R and is not identity.*

This is stronger and cleaner than the original M5 selection argument. It shows that an exotic invariant operator fixing any preassigned R can be constructed without cardinal assumptions. It also sharpens the role of Bacon's rebuilding theorem: the hard part is not constructing the exotic operator, but rebuilding a generic fundamental proposition relative to the enlarged definable stock.

6.7 The T9 and M7 reductions

Theorem 9 (T9 dichotomy). *If PC supplies an injection $\mathcal{P}(K) \rightarrow P$ into the pure unary operators, and every $\approx$ -class injects into G , then*

$$K \text{ is finite} \quad \vee \quad |G| \geq 2^{|K|}.$$

Theorem 10 (Invariant reachability). *Every proposition is reachable from r by an invariant operator if and only if*

$$i \mapsto i \cdot r$$

is injective.

The second theorem turns the open invariant form of Fundamental Completeness into a precise property of the chosen fundamental proposition. It also confirms that the issue depends on Bacon's particular construction rather than following from general facts about substitutions.

7 Where the open question now stands

The current affirmative and negative routes can be stated compactly.

Affirmative route. Construct an E satisfying (SE): E must respect Bacon's substitution action, and every pure unary operator definable using E must already occur among the values $E(n)$. We have explored systematic ways of closing a countable collection of operators while

preserving a proposition that distinguishes its members. Those proposals remain preliminary and are not counted among the verified results in this report.

Negative route. Show that every proposed enumeration omits some pure unary operator definable from that very enumeration. The generalized diagonal D_R identifies one precise route: find a pure relation R that respects actual truth and relates all representations of a suitable proposition $H(q)$. The natural relation generated by bijections fixing q handles representations related by reversible changes of input, but not yet all representations.

Relative plausibility. The formal work has not settled which answer is more likely. It has, however, made an affirmative answer significantly more concrete: it is enough to construct one enumeration satisfying (SE) in the verified substitution model. At the same time, the diagonal restrictions show why the most direct Cantorian objection to such an enumeration fails. A separate positive result now handles every subcollection of Goodman’s principles whose logical-purity instances are confined to identity, negation, constant truth, and constant falsity. The next extension should add purity of K , then purity of conjunction.

8 Remaining verification work on the notes

The following items should not be reported as verified:

1. T7b, until a precise kind-level diagonal formula is supplied.
2. The multiple-fundamental “wide Fun” discussion in M4, which does not state a precise selection condition on Fun or a precise Separated Structure formula.
3. The proposed extension of the verified M5 collision calculation from its displayed operator to merely existentially given invertibles.
4. A full rebuilt version of Bacon’s model containing the repaired exotic operator in its pure stock.
5. Global semantic L2 for Bacon’s complete definable stock. It is currently verified only for identity, necessity, possibility, constant truth, and constant falsity.
6. A model validating PP when the logical-purity schema extends beyond identity, negation, constant truth, and constant falsity. The next cases are K and conjunction; Goodman’s full question requires all closed logical terms at all types.

7. The existence or impossibility of an enumeration respecting Bacon’s substitution action and satisfying (SE).

9 Recommended next steps

We recommend preserving the following lines of attack.

1. **Continue adding logical purity.** Identity, negation, constant truth, and constant falsity are now handled. Add purity of K , then purity of conjunction, while preserving closure of purity under application, Recombination, Exhaustion, Modalized Functionality, and PP. A complete result requires every closed logical term at every type.
2. **Positive construction.** Test whether a countable collection of pure operators can preserve a proposition separating its members while remaining closed under every operation definable in the language.
3. **Sharp negative test.** Search for one closed logical unary F whose action makes two distinct members of the proposed pure collection agree at its separating proposition. Such an F would refute that proposed construction immediately.
4. **Finish the source audit only when formulas are available.** Obtain an exact T7b statement, an exact formulation of the wide-Fun discussion, and an exact statement of the proposed general collision argument before encoding them.
5. **Keep model claims and proof-theoretic claims separate.** In particular, do not promote the conditional M4/M6 stock theorems to claims about Bacon’s exact model until their stock premises are instantiated.

10 Reproducibility

The controlling verification ledger is

`reports/GOODMAN_COMPLETE_VERIFICATION_MATRIX_2026-07-27.md.`

The dedicated audit theory is

`reports/audit_goodman_complete/Audit_Goodman_Complete.thy.`

The final Claude-Codex adversarial transcript is

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reports/CONSENSUS_GOODMAN_FINAL_AUDIT_2026-07-27.md.
```

The project status and handoff files identify the theories corresponding to each result. A successful audit build is obtained from the project root with:

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/opt/homebrew/bin/isabelle build -v -d . \  
-D reports/audit_goodman_complete \  
Goodman_Complete_Audit_2026_07_27
```

The appropriate overall verdict is therefore: *the formalization verifies, corrects, or precisely qualifies every determinate claim in Goodman's notes, including two substantive repairs, and reduces the unrestricted consistency problem to a single explicit model-construction equation. A HOL-ZF model also validates PP and the displayed background together with purity of identity, negation, constant truth, and constant falsity, so every subcollection whose logical-purity instances are confined to those four formulas is consistent. Goodman's question remains open because the intended theory requires purity of every closed logical term at every type.*